

Applied Statistics - Project

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For information, as we said in the notebook, we answered part 4 in all the 3 parts and not in a distinct part. The part 4 of this project is a “bonus” part, not asked directly in the subject but which was interesting in our opinion.

Part 1: Dataset Familiarisation and Descriptive Analysis

Objective

The main goal of this first section is to explore and describe the dataset.

This initial step aims to **understand the basic structure of the data** by looking at its size, key columns, time coverage, and unique identifiers (such as transects, observers, and species).

Next, the goal is to **summarize key variables** (like counts and environmental data) using descriptive statistics, presented in both tables and charts.

We will pay **special attention to the observation effort** and how it changes over time and space. Understanding this is crucial because it directly affects the reliability and interpretation of all future results.

Finally, this section will **identify and discuss any data quality issues**—such as missing values, outliers, or unusual periods—to assess their potential impact on later analyses.

Manual Data Review and Correction

A manual inspection of the data revealed inconsistencies in how transect names were recorded. These issues could have led to the same transect being treated as two different locations in the analysis.

Two specific problems were identified and corrected in the code:

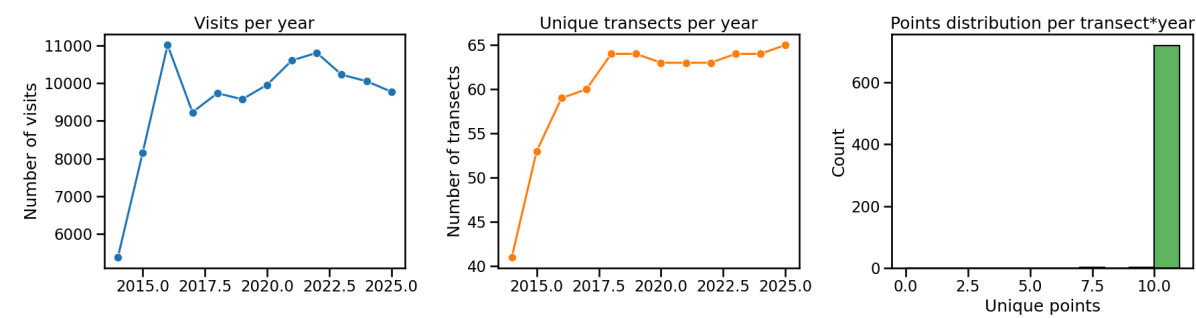
1. **Capitalization Issues:** Some transect names were entered inconsistently, sometimes in uppercase and other times in lowercase, despite referring to the same transect.
2. **Spelling Errors:** A specific misspelling was found where one transect was incorrectly labeled "desmarinière" instead of the correct "desmarinières".

Visualizations

This section presents a series of visualizations created to explore the dataset, focusing on observation effort and the distribution of key categorical variables.

1. Observation Effort Metrics

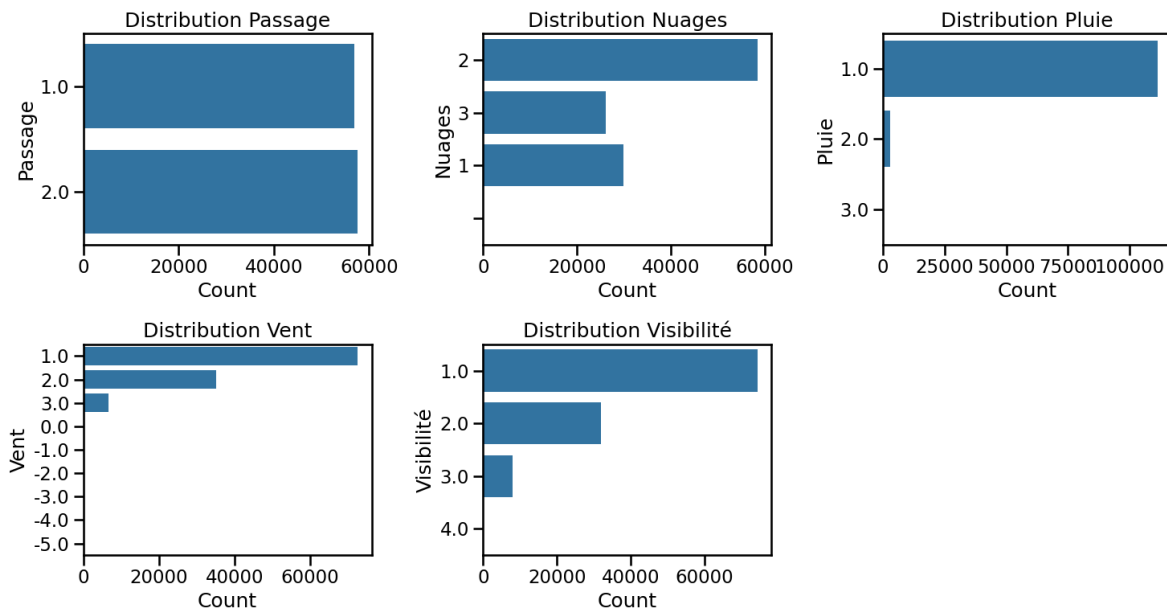
To understand the sampling distribution, we first visualized the observation effort across the study period (2014–2015). These plots help identify patterns, sampling consistency, and potential biases in how data was collected over time and space.



 Visits per year: This line chart shows the total number of observation visits each year. There is a significant peak in 2017 (11,000 visits), followed by a fluctuating but sustained high level of effort (generally between 9,000 and 11,000 visits) in the following years.	Unique transects per year: This chart shows a rapid increase in the number of different transects studied between 2015 (around 41) and 2018 (around 64). After 2018, the number of transects monitored each year remains stable, at around 64-65.	Points distribution per transect*year: This histogram shows that the vast majority of surveys (over 600 occurrences) include exactly 10 unique observation points. This strongly suggests that a standard protocol of 10 points per transect is being applied very consistently.
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2. Categorical Variable Visualization

The second set of visualizations examines the distribution of key categorical data. This includes variables such as weather conditions and visibility.

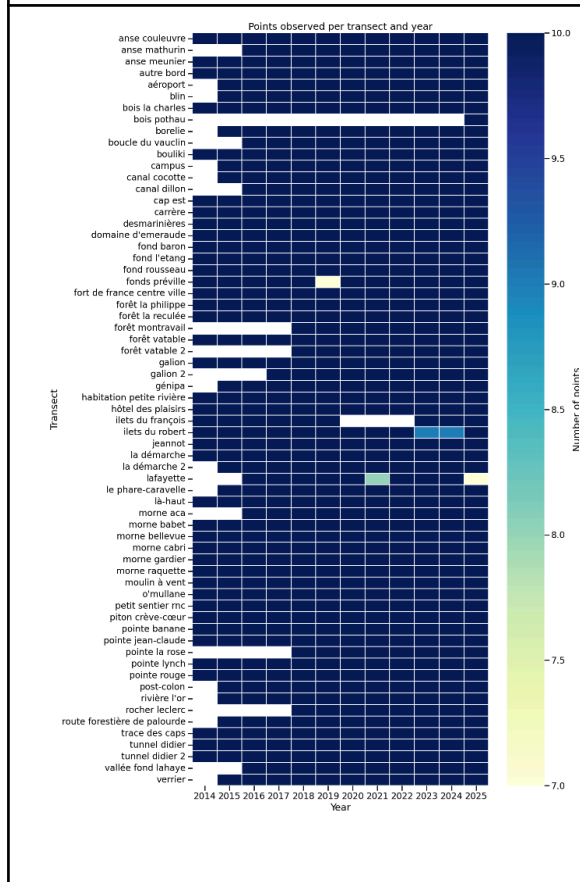


Distribution Passage: This chart shows a very balanced distribution between categories '1.0' and '2.0', each with approximately 55,000 observations. This confirms that the monitoring protocol involves two passages (visits) for each observation site (defined by a transect, point, and year).	Distribution Nuages (Clouds): Category '2' (partly cloudy) is the most frequent condition (around 55,000 observations), followed by category '1' (clear, approx. 30,000) and '3-' (overcast, approx. 25,000). This distribution seems logical, reflecting variable weather conditions where the sky is often neither completely clear nor completely overcast.	Distribution Pluie (Rain): The vast majority of observations (over 100,000) were conducted under condition '1.0' (very likely meaning "no rain"), with observations under other conditions being extremely rare. This suggests that observations are overwhelmingly, or preferably, carried out during dry weather.
Distribution Vent (Wind): Most observations occurred under condition '1.0' (likely low wind), with a count exceeding 60,000, followed by condition '2.0'. Similar to the rain data, this indicates that surveys are preferably conducted in low to moderate wind conditions, likely to standardize the protocol or for comfort reasons.	Distribution Visibilité (Visibility): Category '1.0' (likely "good visibility") is by far the most frequently recorded condition (around 65,000). Logically, since observations are primarily made in dry weather (little rain) and with low wind, it is consistent that visibility is predominantly good during the surveys.	

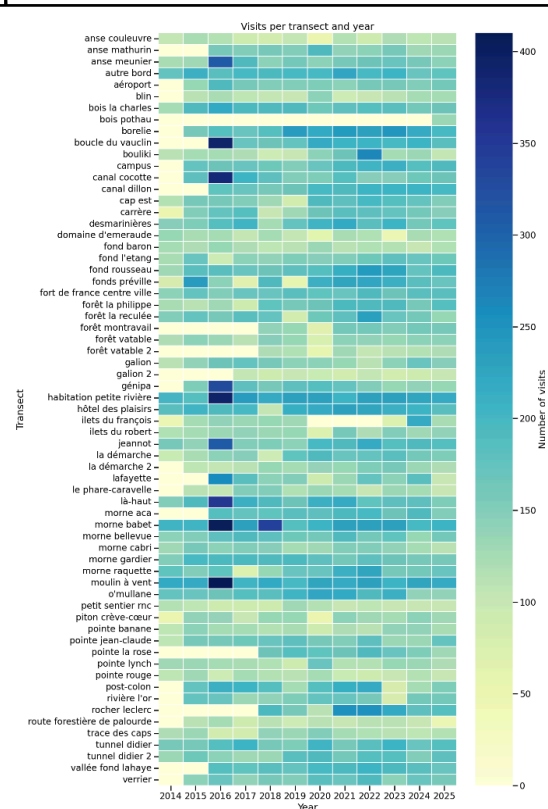
3. Detailed Effort Analysis by Transect

To analyze the distribution and consistency of sampling effort in more detail, two specific heatmaps were generated. These maps cross-reference visits by year and by transect.

The first heatmap shows the **number of unique observation points visited per year and per transect**.



The second heatmap shows the **total number of observations (all points combined) per year and per transect**.



Note: If the graphs in this section are not clearly legible, all figures are available in the figures/part_1 folder.

Conclusion and Recommendations

The exploratory and descriptive analysis of this dataset revealed several inconsistencies as well as heterogeneous sampling effort across time and space.

To prepare a robust dataset for future analyses and to limit interpretation bias, we made the following data cleaning decisions:

Remove all observations from 2014. This year had an overall observation effort (60% below the median) that was too low to allow for reliable comparisons.

Exclude transects that did not have at least 10 observation points visited 6 or more times (identified using the first heatmap: points_by_transect_year).

Remove transects showing a low average number of total visits (identified using the second heatmap).

Recommendations for Future Data Collection

To improve the quality and reliability of data collected in the future, we make the following recommendations:

Standardize Name Entry: It is imperative to avoid unintentional redundancies. Transect names must follow a strict convention:

Unified spelling (e.g., always "desmarinières" and not "desmarinière").

Consistent case (e.g., write everything in lowercase or all in uppercase).

Harmonize Inter-annual Sampling Effort: As much as possible, each transect should be inspected a similar number of times from one year to the next. This is essential for valid temporal comparisons.

Ensure Completeness of Visits: When a transect is visited, effort should be made to inspect *all* observation points associated with that transect, rather than only covering a portion of them.

Failure to follow these recommendations introduces significant risks. Inconsistencies in names or uneven sampling effort create major **observation biases**. These biases can skew results and lead to erroneous interpretations: for example, an apparent decline in a species might merely be an artifact of reduced monitoring effort (fewer visits or fewer points inspected) rather than a true ecological decline.

PART 2: Multi-Year Indicator Trends

1. OUR APPROACH

This section analyzes the monitoring site's global biodiversity trends from 2015 to 2025, providing a "big picture" assessment of the entire bird community's health.

Objective: Establish whether biodiversity indicators show directional change or stability over the 11-year monitoring period. This macro-level analysis provides essential context for Part 3's species-specific findings, as aggregated indicators can mask individual species trajectories—a stable global trend could hide one species declining while another increases.

Our Three-Step Approach:

- Selected four complementary biodiversity indicators to capture different facets of community health (size, variety, structure).
- Quantified uncertainty robustly using bootstrap confidence intervals (95% Confidence Interval (CI), n=400 replications) to distinguish real trends from sampling noise.
- Selected LOESS (Locally Estimated Scatterplot Smoothing) smoothing for trend analysis after careful consideration of linear regression's limitations for ecological data.

We chose non-parametric bootstrap resampling to estimate CIs because it makes no assumptions about how our data is distributed. Ecological count data is notoriously messy—many zeros/NA (species not detected), occasional very high counts (flocking behavior), and everything in between. Traditional statistical methods assume data follows neat patterns (like normal distributions), which rarely holds for bird counts. With n=400 bootstrap replications per year, we generate robust 95% confidence intervals that reflect genuine sampling uncertainty.

Also, we chose LOESS instead of Linear Regression. Linear regression was our initial consideration: it provides simple, interpretable slopes (average change per year) that are easy to communicate. However, especially after visualization we concluded it would be insufficient and potentially misleading for ecological time series:

- **Unrealistic assumption:** Bird communities respond to environmental drivers (habitat change, climate variability) that rarely operate linearly. Even within 11 years, there could be periods of rapid change followed by stability, or cyclical patterns that a straight line cannot capture.
- **High sensitivity to outliers:** With only 11 data points, a single unusual year (such as the previously-removed 2014) can disproportionately skew the entire trend line.
- **Temporal autocorrelation:** Ecological time series show population inertia where consecutive years are more similar than distant years. Linear models treat each year as independent, potentially underestimating uncertainty.

LOESS addresses these concerns. It fits weighted regression at each point using nearby data ($\text{frac}=0.6$), allowing trends to curve naturally without imposing rigid assumptions. It handles small sample sizes gracefully, stays close to the data while smoothing noise, and gives outlier years less global influence. The trade-off is losing a single "slope" value for management communication, but we gain crucial ecological realism—the method follows the data's natural rhythms rather than forcing geometric rigidity.

2. INDICATORS STUDIED

We calculated four indicators annually from the cleaned 2015-2025 dataset:

2.1 Total Abundance

- **Definition:** Sum of all individual birds counted across all species.
- **What it measures:** How many individual birds are present overall—a measure of the ecosystem's ability to support bird populations.
- **Methodology:**
 - For each observation row in the cleaned dataset, we selected a count value using a prioritized hierarchy: "Totaux audio+visuel sans Vol" was preferred as it represents confirmed ground/perch observations; if unavailable, we used "Totaux Audio+visuel avec Vol"; as a final fallback, "Totaux visuel sans Vol" was used. This prioritization ensures consistent count logic across years while maximizing data completeness.
 - Annual totals were computed by summing all preferred count values for each year. To estimate 95% CI, we implemented non-parametric bootstrap resampling ($n=400$ iterations): for each year independently, we resampled the year's observation rows with replacement (maintaining sample size), recalculated the total abundance from the resampled data, and repeated 400 times. The 95% CI bounds are the 2.5th and 97.5th percentiles of the resulting bootstrap distribution.

2.2 Species Richness

- **Definition:** Count of unique species detected per year (presence-based, regardless of abundance).
- **What it measures:** The most fundamental biodiversity metric—how many different species are detected.
- **Methodology:**
 - For each year, we identified all unique species with at least one detection (count > 0) in the cleaned dataset, regardless of how many individuals were observed. Species with zero counts across all observation rows in a given year were excluded. This presence/absence approach makes richness robust to abundance estimation errors and detection heterogeneity—a rare species seen once contributes equally to richness as an abundant species seen hundreds of times.

- Bootstrap 95% CIs were estimated using the same resampling framework as abundance. In our case, they can be asymmetric because rare species detected only once or twice might randomly appear or disappear in bootstrap samples, reflecting genuine uncertainty about whether those detections represent real presence or sampling luck

2.3 Shannon Diversity Index

- **Definition:** $H = -\sum(p_i \times \log(p_i))$, where p_i is relative abundance of species i .
- **What it measures:** Both the number of species and how evenly individuals are distributed among them. A community dominated by one or two species scores lower than one where abundance is spread more evenly.
- **Methodology:**
 - For each year, we first aggregated total counts by species (summing the preferred count column across all observation rows). We then calculated relative abundances: $p_i = (\text{total count of species } i) / (\text{total count of all species})$. The Shannon index was computed by applying the formula $H = -\sum(p_i \times \log(p_i))$, using natural logarithm. Species with zero counts were excluded (as $\log(0)$ is undefined, and by convention, $0 \times \log(0) = 0$).
 - Bootstrap 95% CIs were generated by resampling observation rows within each year ($n=400$ iterations): for each bootstrap sample, we re-aggregated species counts, recalculated relative abundances, and recomputed Shannon's H . Because Shannon is a nonlinear transformation of count data—integrating both richness (number of species with $p_i > 0$) and evenness (distribution of p_i values)—its bootstrap distribution reflects compounded uncertainty from both components.
 - This explains why Shannon CIs are relatively wide (83% of observed data range): uncertainty propagates through the logarithmic transformation, making Shannon more sensitive to sampling variability than simple count metrics.

2.4 Simpson Diversity Index (1-D)

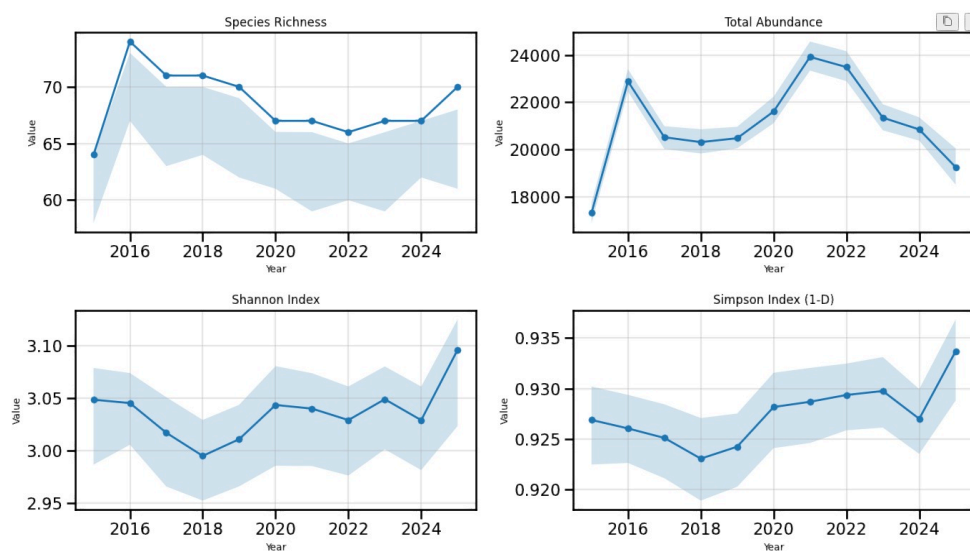
- **Definition:** $1 - \sum(p_i^2)$, representing probability that two randomly drawn individuals belong to different species.
- **What it measures:** The probability that two randomly selected birds belong to different species. Higher values mean more diversity; values near 1 indicate a well-balanced community.
- **Methodology:**
 - Following the same species-by-year aggregation as Shannon, we calculated relative abundances p_i for each year. We then computed Simpson's dominance index $D = \sum(p_i^2)$, which represents the probability that two randomly drawn individuals belong to the same species. The complement ($1 - D$) gives the probability they belong to different species, which scales intuitively: values near 0 indicate extreme dominance (most pairs are the same species), while values near 1 indicate high diversity (most pairs differ).

- Bootstrap 95% CIs used the identical resampling procedure as Shannon. Simpson's quadratic form (summing squared proportions) makes it less sensitive to rare species than Shannon's logarithmic form, but it still exhibits relatively wide CIs because the squaring operation amplifies uncertainty in abundance estimates for dominant species.

Why These Four? They span structural (richness), quantitative (abundance), and compositional (Shannon, Simpson) dimensions, enabling us to distinguish scenarios like: stable richness with changing dominance, abundance fluctuations without compositional shifts, or coordinated changes indicating systemic perturbations.

3. CONCLUSIONS

3.1 Visual Evidence and LOESS Trend Analysis



[Figure 1: Temporal evolution of the four global biodiversity indicators (2015-2025). The shaded area represents the 95% bootstrap confidence interval.]

Species Richness (Top-Left):

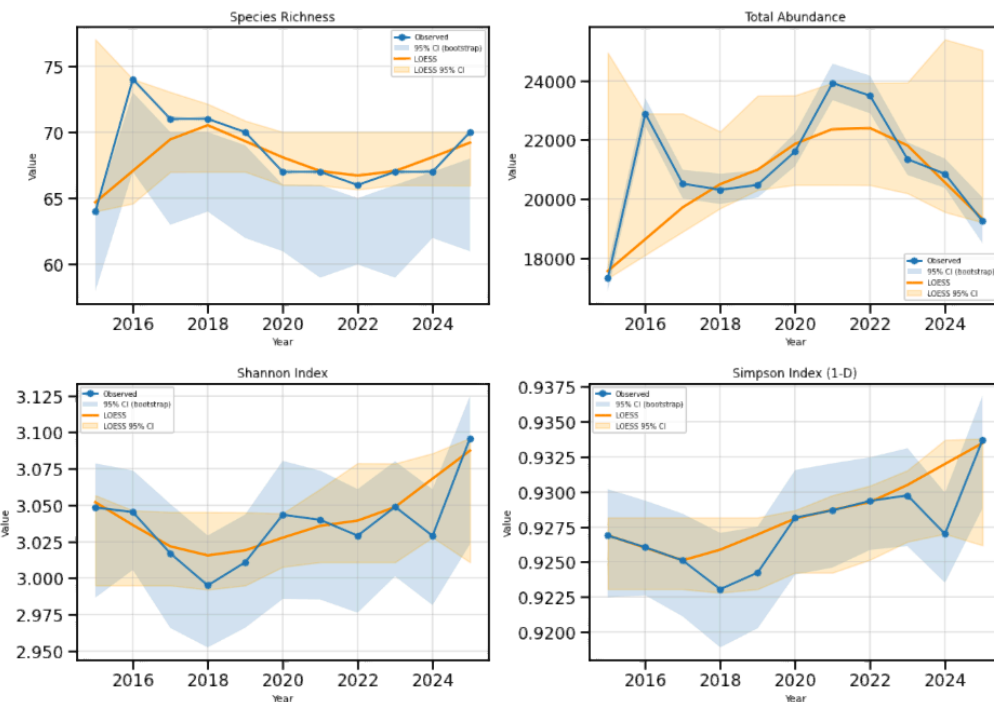
Shows moderate variability, ranging from 64 to 74 species. After an initial peak in 2016, richness exhibits a gentle declining trend stabilizing around 66-70 species. The 95% CI is noticeably wide, indicating a moderate degree of statistical uncertainty in the precise count for any given year. This is an expected result for a richness metric, as the total count is highly sensitive to sampling instability—the random chance of detecting (or missing) a single individual of a rare species during a survey directly impacts the total.

Total Abundance (Top-Right):

Displays the most pronounced variability in observed values, ranging from ~17,000 to ~24,000 individuals. Despite this fluctuation, the confidence interval remains really narrow and close to the line, indicating we are reasonably certain about the abundance estimates each year. The clear hump-shaped pattern (rise, plateau, decline) is well-defined.

Shannon & Simpson Indices (Bottom Row):

Both diversity indices show relatively stable point estimates over time, with Shannon hovering around 3.0-3.10 and Simpson around 0.922-0.934. However, both display relatively wide confidence intervals compared to the magnitude of temporal variation, indicating greater uncertainty in estimating these diversity metrics. Despite this uncertainty, no clear directional trends are evident—community evenness and dominance structure appear stable over the monitoring period—although a substantial increase in both can be seen for 2025.



[Figure 2: LOESS-smoothed temporal evolutions of biodiversity indicators showing observed values (blue), bootstrap 95% CI (blue shade), LOESS curves (orange), and LOESS 95% CI (orange shade)]

Species Richness (Top-Left):

The orange LOESS curve reveals a complex, non-linear pattern. It shows an initial climb from 2015 to a peak around 2018, followed by a shallow dip to a low point around 2022. The trend then appears to flatten and slightly recover toward 2025. The curve ends at a value similar to where it began, suggesting significant fluctuation over the period but no strong, consistent directional trend.

Total Abundance (Top-Right):

The orange curve clearly captures a "boom-and-bust" trajectory. The trend shows a steep rise from 2015, which continues more moderately to a distinct peak around 2021-2022. After this peak, the curve shows a clear and steady decline toward 2024. This single hump-shaped pattern is the dominant feature of the abundance dynamics. The wide LOESS confidence band (orange shading) reflects the high year-to-year volatility of the observed data.

Shannon Index (Bottom-Left):

The orange curve shows a clear "dip-and-recover" pattern, not stability. It displays a noticeable decline from 2015 to a low point around 2018. Following this, the index begins a steady and unambiguous upward trend, accelerating after 2021 and ending at its highest point in 2024. This pattern strongly suggests a shift in the community, with diversity (evenness and richness) clearly increasing in the latter half of the period.

Simpson Index (Bottom-Right):

The orange curve is not flat; it closely mirrors the pattern of the Shannon Index. It shows a slight dip to a minimum around 2018, followed by a consistent and clear upward trend that continues through 2024. This confirms the Simpson diversity is not stable but has been measurably increasing, indicating that the community has become less dominated by any single species in the second half of the study.

3.2 Main Conclusions

a. Community Recovery After 2018 Turning Point

All indicators show synchronized 2018 inflection:

- Richness peaked around 2018 then declined
- Shannon and Simpson hit lowest values around 2018

Post-2018, diversity indices show sustained, accelerating improvement through 2024, reaching maximum values. The biodiversity became progressively more balanced—less dominated by few abundant species, with more even distribution across species. This is a positive biodiversity outcome.

b. Decoupling of Abundance and Diversity (Post-2021)

Abundance and diversity indices moved in opposite directions post-2021:

- Abundance peaked 2021-2022, then declined sharply toward 2024
- Shannon and Simpson continued rising through 2024

Importantly, this abundance decline occurs despite stable or increasing sampling effort (cf. Part 1 'Effort Metrics' plots). This rules out reduced sampling as the driver—the decline probably reflects genuine ecological change, not a sampling artifact. The abundance decline hasn't caused the community to collapse—in fact, diversity kept improving even as total bird numbers fell. This suggests the decline is hitting common/dominant species harder than rare ones, which actually makes the community more balanced overall. The ecosystem isn't failing; it's just reorganizing itself into a new configuration with fewer total birds but a more even distribution across species.

c. Richness Shows Cyclical Fluctuation, Not Directional Trend

Richness fluctuates around stable mean (2016 peak → 2022 dip → 2025 recovery), ending near starting value. This suggests the same core species persist throughout, but their relative dominances are shifting.

d. Justification for Part 3

The abundance decline concurrent with diversity improvement creates compelling need for species-level analysis to answer:

- Which species drove the 2021-2022 abundance peak?
- Which species are declining post-2021?
- Which rare species are increasing post-2018 (improving evenness)?

3.3 Limitations

- Our confidence intervals capture year-to-year variation but don't account for changes in data collection quality over time (e.g., more experienced observers in later years).
- Short monitoring period: 11 years shows us patterns but isn't long enough to determine their causes. We can't tell if trends are driven by weather, habitat changes, or other factors without additional data.
- Abundance going down while diversity goes up could mean the community is healthier (better balanced) OR less productive (losing important species). We need species-level analysis to know which.

3.4 Recommendations

For Management:

- Interpret post-2021 abundance decline cautiously—it coincides with improving diversity, suggesting reorganization not collapse. Effort data suggests this is genuine ecological change, not reduced sampling.
- Investigate 2018 turning point (review records for storms, management actions, habitat changes)
- Monitor whether 2024 diversity peak stabilizes or continues rising
- Do not use abundance alone as health metric—multi-indicator assessment essential

For Future Monitoring:

- Continue recording effort metrics (transects surveyed, observer IDs, survey duration) and enhancing with point-level data (exact survey duration per point, weather conditions, observer experience levels)
- Extend monitoring through 2030 minimum for most robust inference
- Implement effort-adjusted indicators (abundance per transect-hour, occupancy models) using the existing effort data

For Future Analysis:

- Prioritize species-level analysis (Part 3) to explain abundance-diversity divergence
- Stratify by habitat type to test if patterns are uniform or habitat-specific
- Correlate with environmental covariates (weather, storms, disturbance events)
- Decompose diversity into richness vs. evenness components

PART 3: SPECIES-LEVEL ANALYSIS

1. OBJECTIVES AND JUSTIFICATION

Part 3 aims to analyze the temporal evolution of a subset of selected bird species, focusing on their annual counts and presence rates. This analysis allows identification of species-specific patterns that are not visible in global biodiversity indicators (Part 2).

Why this approach?

- Aggregated indicators mask individual species trajectories
- Some species may decline while overall diversity remains stable
- Identifying sentinel or vulnerable species is crucial for conservation
- Understanding factors affecting certain species requires granular analysis

2. SPECIES SELECTION

2.1 Selection Criteria

To ensure diverse ecological representation and statistically robust analyses, we selected species according to three complementary criteria:

1. Top abundance (5 species): The most abundant species over the total period, representing dominant generalists in the community.
2. Habitat specialists (2 species): Species with $\geq 70\%$ of their observations concentrated in a specific habitat, revealing strong ecological dependencies.
3. Detection modality dependence (2 species): One audio-dominant species ($\geq 80\%$ auditory detections) and one visual-dominant ($\leq 20\%$ auditory), allowing assessment of potential detection biases.

Data quality constraints: All selected species have at least 6 years with data and 50 observation rows total, ensuring minimal statistical reliability.

2.2 Selected Species

Nine species were selected based on the criteria above:

Top Abundance Species (5 species - representing dominant generalists):

- Quiscale merle (*Quiscalus lugubris*)
- Sucrier à ventre jaune (*Coereba flaveola*)
- Elénie siffleuse (*Elaenia martinica*)
- Tourterelle à queue carrée (*Zenaida aurita*)
- Sporophile roug gorge (*Loxigilla noctis*)

Habitat Specialists (2 species - with $\geq 70\%$ observations in one habitat):

- Amazone aourou (*Amazona amazonica*) - Urban habitat specialist (90.8% in urban habitat)
- Astrild ondulé (*Estrilda astrild*) - Agricultural habitat specialist (78.5% in agricultural habitat)

Detection Modality Dependence (2 species):

- Colombe rouviolette (*Geotrygon montana*) - Audio-dominant species ($\geq 80\%$ auditory detections)
- Aigrette neigeuse (*Egretta thula*) - Visual-dominant species ($\leq 20\%$ auditory detections)

3. METHODOLOGY

3.1 Calculated Indicators

For each species and each year, we calculated:

1. Annual count: Sum of individuals counted per species and year, using primarily 'Totaux audio+visuel sans Vol' (to exclude flight-only detections, less reliable).
2. Presence rate: Proportion of visits of the year where the species was detected ($\text{count_preferred} > 0$), measuring detection frequency rather than absolute abundance.

3.2 Uncertainty Estimation

95% confidence intervals (95% CI) were obtained via non-parametric bootstrap:

- Resampling with replacement of observation rows within each year ($n=400$ replications)
- Recalculation of indicators for each replication
- Extraction of 2.5% and 97.5% percentiles from the bootstrap distribution

This approach captures uncertainty due to sampling variability without assuming a specific distribution.

3.3 Temporal Trend Estimation

Annual slopes were estimated via simple linear regression on year (centered), with 95% CI by bootstrap of (year, indicator) pairs:

- Slope > 0: average annual increase
- Slope < 0: average annual decrease
- CI excluding 0: statistically discernable trend

4. RESULTS AND ANALYSES

4.1 Figure 1: Individual Species Evolution (Count and Presence) (too big for this document, look at figures/part3/species_all_comparison_counts_presence.png)

This figure shows: Two panels per species (count on left, presence on right) with 95% CI.

Main observations:

Ecological Groups

Generalists (Top abundance - Quiscale merle, Sucrier à ventre jaune, Elénie siffleuse, Tourterelle à queue carrée, Sporophile rougegorge):

- High counts: 1500-4000 individuals/year depending on species
- Very stable presence: ≥99% after 2016, reflecting quasi-systematic detection
- Narrow CI: Well-estimated populations, low uncertainty
- Common temporal pattern: Increase 2014-2016, then stabilization or moderate fluctuations
- Notably, Quiscale merle dominates with the highest counts (3000-3500 individuals/year)

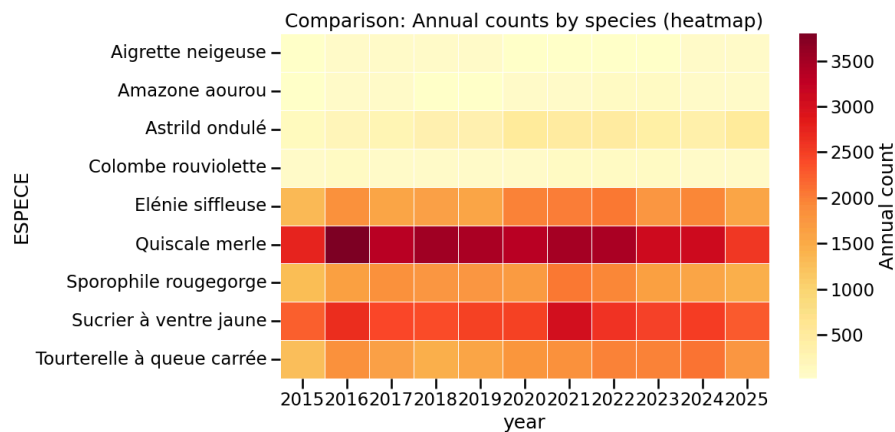
Habitat specialists (Amazone aourou, Astrild ondulé):

- Low counts: 50-550 individuals/year, but generally increasing
- Variable presence: 90-100%, with occasional dips suggesting sensitivity to conditions
- Wide CI: More uncertainty related to smaller populations
- High temporal variability: Sensitive to habitat disturbances
- Amazone aourou shows strong urban habitat specialization (90.8% in urban habitat)
- Astrild ondulé shows agricultural habitat specialization (78.5% in agricultural habitat)

Modality-dependent species:

- Colombe rouviolette (audio-dominant): Moderate counts (70-90/year), very stable presence (100%), reflecting reliable auditory detection
- Aigrette neigeuse (visual-dominant): Low counts (20-50/year), most variable presence (85-100%), illustrating sensitivity of visual detection to observation conditions (weather, visibility)

4.2 Figure 2: Annual Counts Heatmap

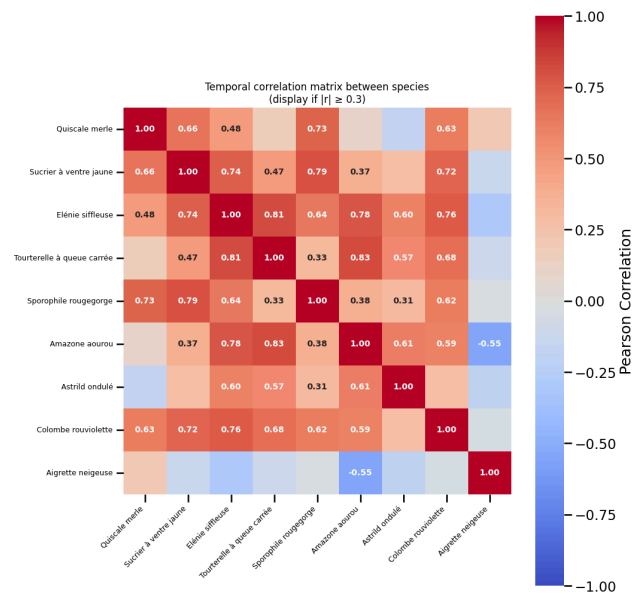


This figure shows: A species × years matrix with color codes for abundance levels.

Main observations:

1. Clear abundance hierarchy: Quiscale merle dominates with counts of 3000-3500/year, followed by Sucrier à ventre jaune and Sporophile rougegorge (1500-2500/year), then moderate species (1000-2000/year) and rare species (<500/year)
2. General temporal pattern: Increase 2014-2016, post-2016 stabilization for dominants
3. Variability: Rare species (Amazone aourou, Astrild ondulé, Aigrette neigeuse) show more marked fluctuations (CI suggesting more uncertainty)
4. Typical abundance distribution: A few very common species (Quiscale merle, Sucrier), several moderate (Elénie, Tourterelle, Sporophile), several rare (specialists and modality-dependent) - classic ecological community structure

4.3 Figure 3: Temporal Correlation Matrix



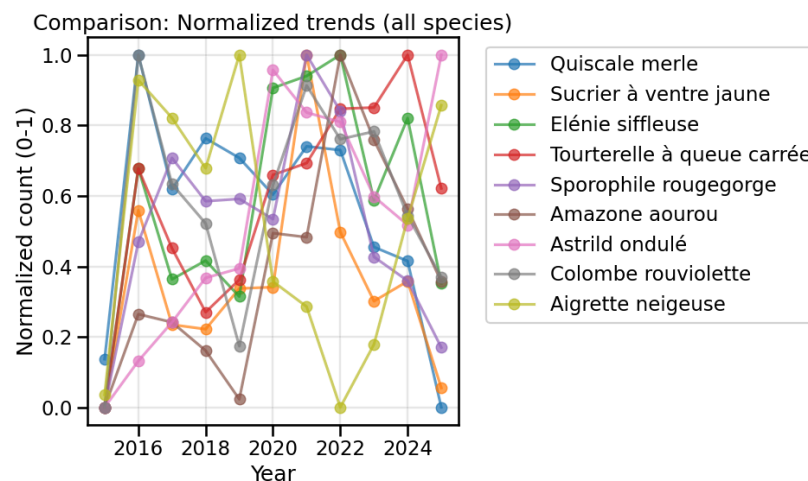
This figure shows: Pearson correlations between species temporal series.

Main observations:

1. High correlation cluster ($r > 0.85$): Quiscale merle, Sucrier à ventre jaune, Elénie siffleuse, Tourterelle à queue carrée, Sporophile rouggorge, and Colombe rouviolette show highly synchronized trajectories
2. Ecological synchronization: These six species likely share common environmental factors (effort, climate, resources), suggesting that their population dynamics are driven by similar underlying processes
3. Aigrette neigeuse isolated: Moderate correlations (0.42-0.64) with the cluster, very weak (<0.3) with specialists - decoupling confirmed. This isolation reinforces the hypothesis of visual detection bias affecting this species differently from auditory-dominant species.
4. Specialists moderately correlated: Amazone aourou and Astrild ondulé show correlations of 0.79-0.81 with the cluster - partial synchronization, but modulated by habitat-specific factors

Interpretation: Synchronization suggests that effort or global condition variations affect most species consistently. Isolation of Aigrette neigeuse reinforces the hypothesis of visual detection bias.

4.4 Figure 4: Overlaid Normalized Trends



This figure shows: All species normalized (0-1) on the same scale to compare trend shapes.

Main observations:

Synchronized temporal patterns:

- 2015-2016: Rapid generalized increase- observed across all generalist species
- 2016-2020: Synchronized fluctuations around high values - all species follow similar patterns
- 2020: Synchronized peak - particularly favorable year for several species
- 2020-2022: Synchronized generalized decline - probable major environmental event (cyclone, habitat disturbance?) affecting all synchronized species
- 2022-2025: Partial recovery, stabilization at levels below previous peaks

Sentinel species:

- Sporophile rougegorge shows the most marked variation amplitude - identified as potential sentinel species to detect environmental perturbations early. Its normalized trend shows the greatest amplitude of variation, making it highly sensitive to environmental changes.
- Aigrette neigeuse decoupled - independent trajectory, requires separate analysis due to visual detection modality bias

Event 2020-2022: Synchronized decline suggests a common factor (climate, habitat, effort).

Recommended investigation: check major weather events, habitat disturbances, sampling effort changes during this period.

4.5 Temporal Trend Slopes Analysis

This analysis quantifies the annual rate of change (slope) for each species, with 95% confidence intervals.

Count trends:

- Tourterelle à queue carrée shows the strongest positive trend (+53.9 individuals/year), with CI [14.1, 94.6] excluding 0, indicating a statistically significant increase

- Elénie siffleuse shows a positive trend (+32.0 individuals/year), though CI [-10.9, 85.5] includes 0, suggesting uncertainty
- Astrild ondulé shows a significant positive trend (+30.3 individuals/year), CI [15.7, 46.2] excluding 0
- Quiscale merle shows a negative trend (-39.6 individuals/year), but CI [-120.7, 57.6] includes 0, indicating high uncertainty
- Most generalist species show trends with wide confidence intervals that include 0, reflecting uncertainty about true trend direction
- Amazone aourou shows a small but significant positive trend (+5.3 individuals/year), CI [2.1, 10.5] excluding 0
- Aigrette neigeuse shows a near-zero trend (-0.4 individuals/year), CI [-3.5, 1.4], indicating relative stability

Presence trends:

- All species show very small positive slopes for presence (<0.01 per year), reflecting that presence rates are already near maximum (90-100%)
- Amazone aourou shows the largest presence slope (+0.0097/year), but still very small
- Most CI for presence slopes include 0, indicating no statistically significant changes in detection frequency
- This is expected given that most species are already detected in >95% of visits

Interpretation:

- Positive count trends for Tourterelle, Elénie, and Astrild suggest these species are expanding or recovering
- Wide confidence intervals for most species reflect high interannual variability and limited temporal coverage
- The lack of significant presence trends confirms that detection frequency is stable, suggesting that observed count trends reflect real population changes rather than detection bias
- Specialists (Amazone aourou, Astrild ondulé) show positive trends, suggesting potential recovery or expansion despite their small populations

4.7 Statistical Tests and Group Comparisons

Group comparisons:

- Mean abundance by group: Top abundance species show mean ~2189 individuals (SD ~686), while habitat specialists show mean ~208 individuals (SD ~220)
- This confirms the ecological trade-off: generalists are 10× more abundant than specialists
- Modality-dependent species show intermediate values, with audio-dominant (Colombe rouviolette) showing higher abundance than visual-dominant (Aigrette neigeuse)

Trend slopes by group:

- Top abundance species show mixed trends: some positive (Elénie, Tourterelle), some negative (Quiscale), with high variability
- Habitat specialists show positive trends overall, suggesting potential recovery
- The wide variation in trends within groups suggests species-specific factors beyond group membership

Key findings:

- Quiscale merle has the highest mean abundance (3258) but shows a negative trend, suggesting potential concern despite high current abundance

- Aigrette neigeuse has the lowest mean abundance (33) and lowest presence rate (95.2%), confirming its status as the most vulnerable species
- Amazone aourou shows the highest variability (CV=0.48) but also a positive trend, suggesting recovery potential despite high sensitivity
- Tourterelle à queue carrée shows both high abundance and strong positive trend, indicating a successful generalist species

5. LIMITATIONS AND METHODOLOGICAL CONSIDERATIONS

1. Imperfect detectability: Presence based on detection>0 does not correct for imperfect detectability (no occupancy model). Cryptic or discrete species may be underestimated.
2. Modality detection bias: Aigrette neigeuse (visual-dependent species) clearly shows bias related to observation conditions. Recommendation: stratified analyses by weather conditions for visual-dependent species.
3. Small populations: Rare species (Amazone aourou, Astrild ondulé, Aigrette neigeuse) show wide CI - less precise estimates. Need to increase effort or monitoring duration for these species.
4. Spatial scale effect: Analysis aggregated at island scale. Local losses/gains may compensate, masking important regional patterns.

6. CONCLUSIONS AND RECOMMENDATIONS

6.1 Main Findings

1. Strong temporal synchronization: Generalist species (Quiscale merle, Sucrier à ventre jaune, Elénie siffleuse, Tourterelle à queue carrée, Sporophile rougegorge) follow very similar trajectories, suggesting dominant common environmental factors affecting these species similarly.
2. Specialist vulnerability: Specialized species (Amazone aourou, Astrild ondulé) show high variability and increased sensitivity - require particular attention for conservation due to their habitat-specific dependencies.
3. Visual detection bias: Aigrette neigeuse (visual modality-dependent species) isolated by different signal, requires specific methodological treatment. Its decoupling from other species suggests that visual detection is more sensitive to observation conditions than auditory detection.
4. Event 2020-2022: Major synchronized decline followed by partial recovery - deserves in-depth investigation to identify cause(s). This event affected all synchronized species simultaneously, suggesting a common environmental driver.
5. Sentinel species: Sporophile rougegorge serves as an indicator of environmental perturbations, showing the most marked variation amplitude in normalized trends. This species could be used as an early warning system for detecting environmental disturbances.

6.2 Recommendations

For management:

- Monitor habitat specialists (Amazone aourou, Astrild ondulé) particularly - more fragile populations requiring specific habitat protection
- Protect specific habitats used by specialists: urban habitats for Amazone aourou, agricultural habitats for Astrild ondulé
- Standardize observation conditions to reduce visual detection bias, especially for Aigrette neigeuse

For monitoring:

- Document major environmental events (weather, disturbances) to explain temporal patterns, particularly the 2020-2022 decline event
- Increase sampling effort for rare species (specialists and visual-dependent species) to improve precision
- Stratify analyses by habitat and weather conditions to identify specific factors affecting different species groups
- Use Sporophile rougegorge as sentinel species to detect early perturbations due to its high sensitivity to environmental changes

For research:

- Investigate causes of 2020-2022 decline (weather events, habitat changes, effort) affecting all synchronized species
- Model detectability to correct biases (occupancy models), particularly for visual-dependent species like Aigrette neigeuse
- Analyze synchronization with environmental variables (precipitation, temperature, extreme events) to understand common drivers affecting generalist species

PART 4: WEATHER EFFECTS ANALYSIS (Bonus)

1. OBJECTIVES AND JUSTIFICATION

Throughout Parts 2 and 3, we observed patterns suggesting that meteorological conditions might influence bird observations. In Part 2, synchronized fluctuations in biodiversity indicators (particularly the 2020-2022 decline event) raised the hypothesis that weather anomalies could partially explain these patterns. In Part 3, species-specific analyses revealed high variability, with Aigrette neigeuse (a visual-dependent species) showing patterns suggesting sensitivity to observation conditions. The recommendation to "stratify analyses by weather conditions" emerged directly from Part 3's findings.

Part 4 provides a focused analysis of how meteorological conditions (rainfall and cloud cover) affect bird observations. This analysis helps interpret temporal trends and identify potential detection biases, though we note that comprehensive climate data is not available to fully explain long-term ecological patterns.

Available data: Rainfall and cloud cover recorded on categorical scales (1-3) during each observation visit. Limitations: Weather scales are coarse, and we lack comprehensive climate data (temperature,

humidity, seasonal patterns) to correlate with long-term trends. This analysis focuses on detection-level effects (how weather during a visit affects observations) rather than climate-level effects (how long-term climate drives population trends).

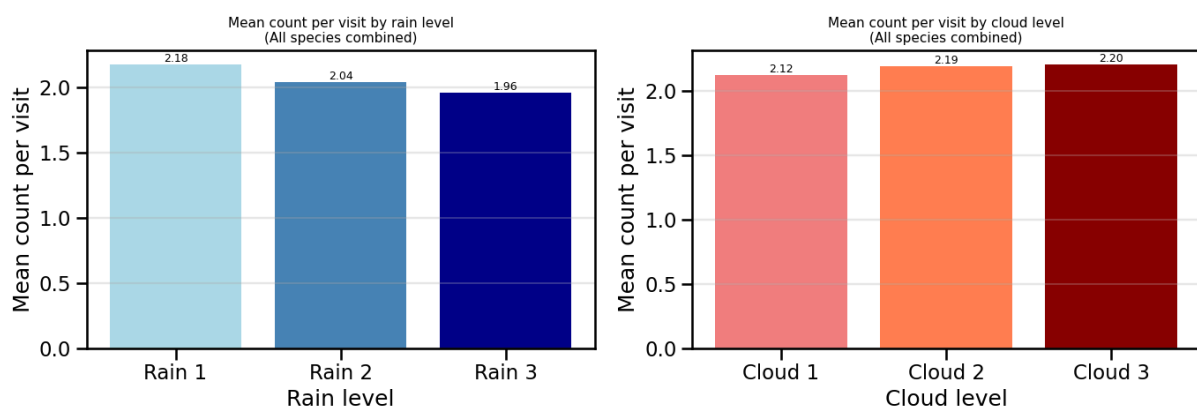
2. METHODOLOGY

We analyzed 108,984 observations (2015-2025) with valid weather data across 102 species. The analysis has two levels:

1. Global effects: Mean counts per visit by weather level (rain 1-3, cloud 1-3) across all species.
2. Species-specific sensitivity: Kruskal-Wallis tests to identify species with significant weather-dependent variation ($p < 0.05$), requiring at least 2 weather levels with ≥ 10 visits per level.

3. RESULTS

3.1 Global Weather Effects



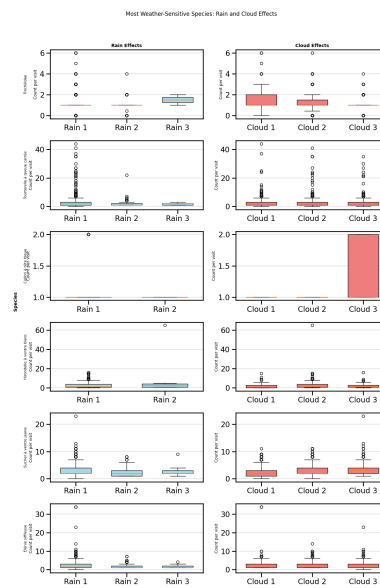
[Figure 1: Mean count per visit by rain level (left) and cloud level (right), all species combined]

Rainfall: Mean counts decrease from 2.18 (Rain 1, dry) to 1.96 (Rain 3, heavy rain), representing a ~10% reduction. Rain systematically reduces observations, likely due to reduced bird activity (seeking shelter) and/or harder detection (reduced visibility, suppressed vocalizations).

Cloud cover: Mean counts increase from 2.12 (Cloud 1, clear) to 2.20 (Cloud 3, overcast), representing a ~4% increase. Cloud cover slightly improves observations, possibly due to better foraging conditions or improved detection (less glare, better contrast).

Key finding: Rain has a stronger negative impact (10% reduction) than clouds have a positive impact (4% increase). Both effects are consistent and could bias trend estimates if weather patterns vary between years.

3.2 Species-Specific Sensitivity



[Figure 2: Most weather-sensitive species showing rain effects (left column) and cloud effects (right column) for 6 species]. To have a better visualization of the graph, you can go in [figures/part_4/'weather_sensitive_species.png](#)

Statistical summary: Among 102 species analyzed, ~31 had sufficient data for rain testing and ~57 for cloud testing. 6 species (6% of all species, 19% of testable species) show significant rain sensitivity, and 4 species (4% of all species, 7% of testable species) show significant cloud sensitivity.

Most rain-sensitive species:

1. Tourterelle à queue carrée: Dramatic decrease (median ~5 at Rain 1 → ~0-1 at Rain 2-3). Strongest rain sensitivity in the dataset.
2. Sucrier à ventre jaune: Moderate decrease (median ~4 → ~2-3)
3. Elénie siffleuse, Sporophile roug gorge, Colombe rouviolette: Moderate to slight sensitivity

Most cloud-sensitive species:

1. Sucrier à ventre jaune and Elénie siffleuse: Slight increase with cloud cover (prefer overcast conditions)
2. Trochilidae (Colibris): Decrease with heavy cloud cover (prefer clear skies for thermoregulation)

Key patterns: Tourterelle à queue carrée shows the most striking rain sensitivity, with observations dropping dramatically during rain. Trochilidae prefer clear skies (thermoregulation needs), while Sucrier and Elénie show slight preferences for overcast conditions.

4. IMPLICATIONS FOR PARTS 2 AND 3

Part 2 connections: The synchronized fluctuations in biodiversity indicators (particularly the 2020-2022 decline) could potentially be partially explained by weather patterns, though we lack comprehensive climate data to test this directly. Weather could contribute to the wide confidence intervals in diversity indices and the high interannual variability in abundance estimates. However, the post-2021

abundance decline concurrent with diversity improvement is unlikely to be explained by weather alone, as weather effects are relatively consistent across species.

Part 3 connections: The high temporal correlations between generalist species ($r > 0.85$) could reflect, in part, shared responses to weather conditions, though the synchronization is too strong to be explained solely by weather. Aigrette neigeuse, identified in Part 3 as isolated with visual detection bias, shows weather sensitivity in Part 4 but less pronounced than expected, suggesting other factors (observer experience, time of day) also contribute. Sporophile rougegorge, identified as a sentinel species in Part 3, shows moderate weather sensitivity, suggesting its high variability reflects both weather effects and genuine environmental sensitivity.

5. LIMITATIONS

1. Coarse weather scales (1-3) may not capture fine-grained differences
2. Limited climate data prevents determining if weather drives population trends or only affects detection
3. Cannot distinguish between behavioral changes, detection biases, and true population responses
4. Many species have insufficient data across weather levels for reliable testing
5. Weather may be correlated with other unmeasured factors

6. CONCLUSIONS AND RECOMMENDATIONS

Main findings:

1. Weather has measurable effects: Rain reduces observations by ~10%, clouds increase by ~4%
2. ~6% of species show significant rain sensitivity, ~4% show cloud sensitivity
3. Weather effects are moderate but meaningful—could bias trend estimates if weather patterns vary between years
4. However, weather alone cannot explain major patterns from Parts 2 and 3 (e.g., 2020-2022 decline, abundance-diversity decoupling)

Recommendations:

- For data interpretation: Account for weather when comparing observations across years/sites; consider weather-adjusted trends for sensitive species
- For survey planning: Standardize weather conditions when possible; avoid heavy rain for sensitive species like Tourterelle à queue carrée
- For future monitoring: Record detailed weather data (continuous measures); link to regional climate data to distinguish detection from population effects

Integration: Part 4 provides essential context for interpreting Parts 2 and 3, but cannot fully explain the major patterns observed. Weather creates measurable detection-level biases, but the major ecological patterns (community recovery post-2018, abundance-diversity decoupling, species synchronization) likely reflect genuine ecological processes rather than weather artifacts. Accounting for weather effects can improve the precision of trend estimates, especially for weather-sensitive species.